Architecture and Training Design Under a Hard Parameter Budget: A Controlled Ablation Study at the 16 MB Scale

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Abstract

We study how architectural and training-recipe choices interact with post-training quantization for decoder-only language models trained from scratch under a 16 MB artifact constraint, the regime defined by the OpenAI Model Craft: Parameter Golf challenge [1]. We run 46 controlled 20,000-step experiments on FineWeb [2] that vary one design axis at a time – quantization precision (INT8 / INT6 / TurboQuant-INT4), positional encoding (full / partial / learned / none), optimizer (Muon / AdamW / Adafactor), weight parameterization (dense / low-rank / block-diagonal / random projection), attention configuration (Multi-Query Attention), and learning-rate / warmup schedule – with three-seed replication on the surviving in-budget axes. The headline finding is the resolution of the central tension left open by our progress report: capacity expansion (more layers, wider MLP) genuinely improves training-time loss, but *only* INT8 post-training quantization with per-row symmetric scales carries that gain through the round-trip. INT6 QAT loses 0.12 bpb; TurboQuant-INT4 [6] loses 2.44 bpb. Reframed as a size-quality Pareto frontier, our best in-budget configuration (partial RoPE, 9 layers, INT8) reaches **1.2147 bpb** at 15.87 MB, and our best over-budget configuration (INT8, 11 layers) reaches **1.1944 bpb** at 19.30 MB – 0.030 bpb below the 1.2244 challenge reference at a 3.3 MB budget overage.

Keywords

language models, quantization, ablation study, parameter-efficient training, Pareto frontier

1 Introduction and Framing

The proposal posed three primary research questions (quantization precision, positional encoding, optimizer) and one exploratory question (parameter-efficient weight factorizations). The progress report expanded the design space with a stacked-configuration experiment (O5) that produced a partial answer: a wider/deeper architecture under INT6 quantization had lower training-time loss but substantially higher post-quantization loss; the two effects nearly cancelled. The progress report queued two follow-ups – a TurboQuant-INT4 rerun of the expanded architecture, and an INT8 rerun at the next-smallest expansion (11 layers, MLP multiplier 2). Both are now complete and together resolve the question.

Beyond resolving O5, this final report adds three new axes that the progress report did not cover: Multi-Query Attention [7] (O6), QK-gain initialization (O7), and the eval protocol itself (O9). It also reports a 9-cell learning-rate / warmup sweep on the in-budget winner (O8, null result), and three-seed replication on the surviving in-budget axes.

Table 1: Baseline architecture and training hyperparameters.

Component	Symbol	Baseline value
Layers	L	9
Hidden dimension	d_{model}	512
Attention heads	h	8
KV heads (GQA)	h_{kv}	4
MLP expansion	m	2 (squared-ReLU)
Context length	ℓ_{ctx}	1024
Vocabulary	$\lvert V\rvert$	1024
Tied embeddings	–	input = output
Training steps	T	20,000
Optimizer (baseline)	–	Muon [5] + Adam
Quantization (baseline)	–	INT8 PTQ, per-row scales

The framing has also changed since the progress report. We now treat the 16 MB constraint as a *reference*, not a binding constraint. Several configurations exceed 16 MB by a few megabytes while delivering clearly lower bpb; under a size-quality Pareto framing those points are part of the legitimate finding. The strict-in-budget result is reported as a separate row, and the headline figure of the paper (Figure 10) is the Pareto curve itself.

2 Experimental Setup

2.1 Dataset and tokenization

All models are trained from scratch on FineWeb [2], a filtered, deduplicated Common Crawl derivative. Training uses the 10 billion-token shard (`fineweb10B`) and the 1,024-token SentencePiece BPE tokenizer (`sp1024`) adopted by the challenge. Input/output embeddings are tied throughout. Training reads contiguous token streams of length ℓ_{ctx}=1024 from pre-shuffled, pre-tokenized `uint16` binary shards. Total training exposure per condition is approximately 1.05×10^{10} tokens (20,000 steps × 8 micro-batches × 2 GPUs × 1024 tokens, adjusted for accumulation).

2.2 Baseline architecture

The baseline (Table 1) is a 9-layer decoder-only transformer with d_{model}=512, 8 attention heads, 4 KV heads (grouped-query attention), squared-ReLU MLP at expansion 2, and partial RoPE applied to the first 32 dimensions of each head. Single-axis ablations mutate exactly one row from this baseline; the stacked configurations in O5 mutate several rows jointly.

2.3 Quantization variants

We evaluate four post-training quantizers, applied to the same training recipe. The first three were specified in the proposal; the fourth (TurboQuant) was added in response to the INT6 fragility finding from the progress report (Section 3, O1).

- **INT8 PTQ** – 8-bit post-training quantization with per-row symmetric scales. This is the challenge's published baseline pipeline.
- **INT6 QAT+PTQ** – 6-bit quantization with quantization-aware training using straight-through estimators [3]. Introduced because INT6 triples effective parameter count within the 16 MB budget relative to FP32, making it nominally the highest-leverage precision lever.
- **TurboQuant-INT4** [6] – a random-rotation + Lloyd–Max scalar quantizer added in response to O1. Rotating each weight row by a random orthogonal matrix concentrates its distribution to a known Beta law, allowing a non-uniform scalar codebook designed for that law to achieve provably near-optimal MSE at very low bit widths. TurboQuant reports quality neutrality at 3.5 bits on LLaMA KV-cache quantization; we apply the same machinery to model *weights* at 4 bits via the pure-PyTorch `pyturboquant` reference implementation. The progress report committed to this axis as the planned mitigation for INT6's quantization fragility; the final result (Section 4) falsifies that mitigation.
- **BF16** – bfloat16 with no quantization, used as the in-memory training reference and for the `val/bpb` column in our results.

2.4 Artifact pipeline

The 16 MB reference applies to the post-quantization, `zlib`-compressed artifact plus training code. To mirror the challenge evaluation, every reported run executes the following pipeline at the end of training: (1) train in bfloat16 (or with QAT for INT6/INT4); (2) quantize weights to the target precision with per-row symmetric scales, keeping 1D tensors (biases, norms) in bfloat16 to avoid disproportionate error; (3) serialize the quantized state-dict and `zlib.compress` it – this is the artifact measured against 16 MB; (4) decompress, dequantize, rehydrate the model, and re-run validation. Two distinct bpb numbers therefore exist for every run: `val/bpb` (training-time, in-memory parameters) and `final/val_bpb` (post round-trip, leaderboard-equivalent). Their difference is the quantization gap Δ_q.

3 Single-Axis Ablations: O1–O4

3.1 O1 – Quantization precision

INT8 PTQ with per-row symmetric scales incurs Δ_q=0.008 bpb. INT6 QAT under the same scheme reaches the same in-memory loss but incurs Δ_q=0.141 bpb under round-trip (Figure 1). The straight-through estimator successfully matches INT8 in-memory, but uniform 6-bit scalar quantization cannot represent the post-training weight distribution without appreciable error. This contradicts the optimistic prediction from existing precision scaling laws [3] in the sub-21M parameter regime, and reframes INT6 as a capacity multiplier that is only useful if a better quantizer is supplied – the setup that motivated TurboQuant (Section 2.3).

3.2 O2 – Positional encoding

Four schemes were compared under otherwise-identical settings (Figure 2). Ordering at 20,000 steps: partial RoPE < full RoPE < learned absolute ≪ none. Partial RoPE at 1.2147 bpb final is the best in-budget condition in the entire study. The gap from partial RoPE to no-PE (0.054 bpb) dwarfs the gap between any two position-aware schemes (≤0.020 bpb): absolute position information matters at this scale; the form of the prior is a second-order variable.

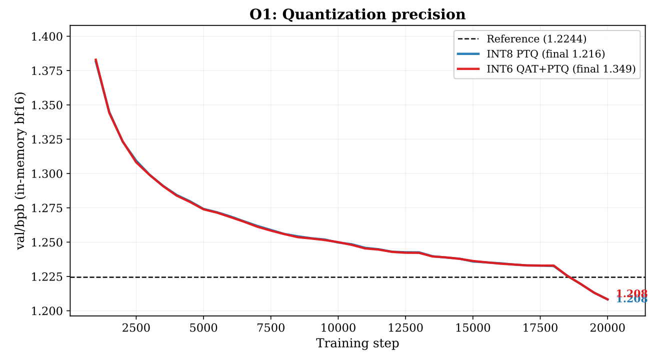

Figure 1: O1 – in-memory bpb across training. INT6 and INT8 reach identical in-memory loss; the divergence in the post-quant round-trip is therefore attributable to the quantizer alone, not to training dynamics.

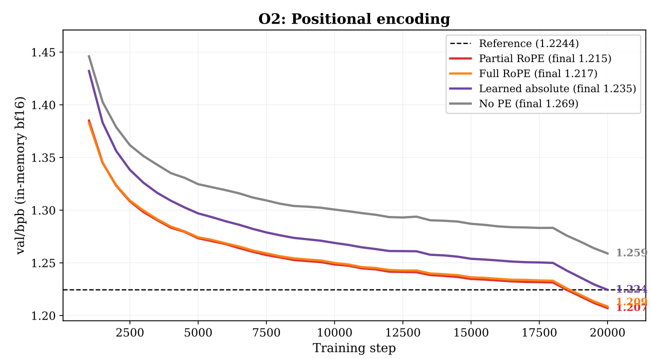

Figure 2: O2 – the three position-aware schemes converge to within 0.02 bpb of each other; no-PE trails clearly. Partial RoPE lowest.

3.3 O3 – Optimizer

Muon – Nesterov momentum applied along the Newton–Schulz-orthogonalized gradient of each 2D weight matrix [5] – reaches 1.2168 bpb final and is indistinguishable from the challenge-reference INT8 baseline. AdamW at tuned learning rates plateaus more than 0.2 bpb higher; Adafactor's factorized second moment loses even more (Figure 3). At under 5M trainable parameters, Adafactor's optimizer-state savings do not compensate for its loss penalty, and AdamW's per-parameter second moment provides no advantage over Muon's orthogonalization.

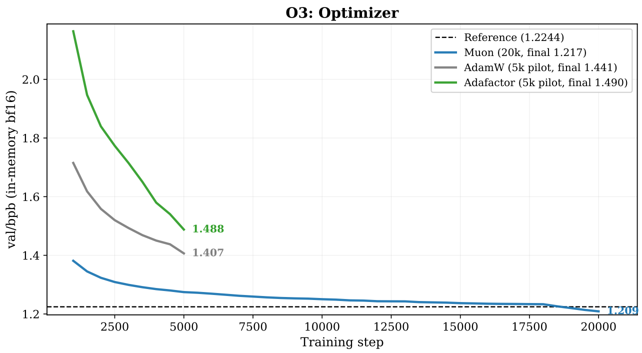

Figure 3: O3 – Muon converges to a qualitatively different loss band than either AdamW or Adafactor at this scale.

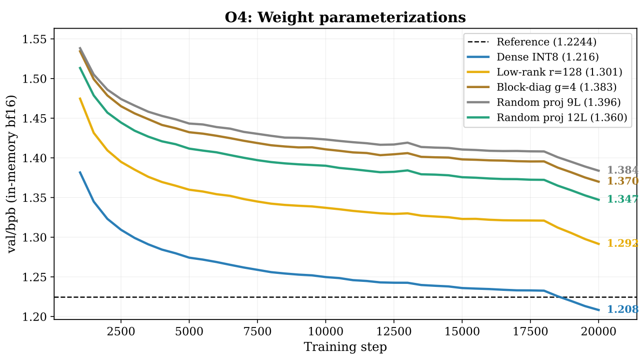

Figure 4: O4 – weight-parametrization variants all converge above the dense INT8 baseline regardless of the structural prior.

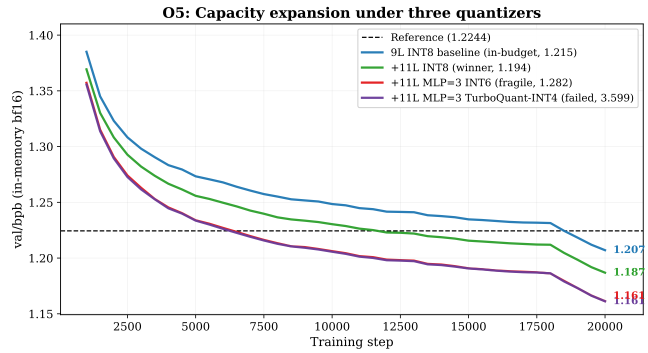

Figure 5: O5 resolved – in-memory bpb during training for the four configurations. INT6 and TQ4 curves overlap (same architecture; only the post-training quantizer differs). The training-time ranking does not predict the round-trip ranking: TurboQuant-INT4 lands at 3.6 bpb final despite reaching 1.16 bpb in-memory.

3.4 O4 – Parameter-efficient weight factorizations

All three structured weight parametrizations (low-rank r=128, block-diagonal g=4, fixed random projection k=128) landed between 1.30 and 1.40 bpb final, cleanly worse than the dense INT8 baseline at 1.2161 (Figure 4). Random projection was retested at 12 layers to rule out capacity starvation; it improved to 1.3601 but still failed to clear baseline. O4 is therefore reported as a falsified hypothesis: under this budget and token horizon, structured weight factorizations do not compete with dense matrices.

4 O5 Resolved: Capacity vs. Quantization

The progress report observed that combining O1–O3's winners with a wider/deeper architecture (partial RoPE + INT6 + 11 layers + MLP=3) reached 1.1614 bpb in training – 0.063 bpb below the 1.2244 reference – but degraded to 1.2816 bpb post-quantization, forfeiting the training-time gain. The two queued follow-ups attempt to recover the gap from orthogonal directions: TurboQuant-INT4 holds architecture fixed and varies the quantizer; INT8 + 11L + MLP=2 holds the quantizer fixed and varies the architecture.

TurboQuant-INT4 fails catastrophically. The same architecture that reached 1.1614 bpb in-memory under INT6 reaches the same in-memory loss under TurboQuant-INT4 (curves overlap in Figure 5), but the post-quantization round-trip yields **3.5991 bpb** – a Δ_q of 2.44 bpb, sixteen times larger than INT6's already-fragile penalty. This is consistent with TurboQuant's published support being concentrated on KV-cache (activations) rather than weights; on the post-training weight distribution at our scale, four bits of precision sits below the representable threshold even with random rotation and Lloyd–Max codebooks.

INT8 + 11L + MLP=2 succeeds cleanly. The conservative architectural expansion under the known-safe quantizer reaches 1.1869 bpb in-memory and 1.1944 bpb post-quantization, a 0.030 bpb improvement on the 1.2244 reference and a 0.022 bpb improvement on the in-budget partial-RoPE winner. Artifact size is 19.30 MB, a 3.3 MB overage on the 16 MB reference.

The combined evidence isolates which lever is binding at this budget: it is the quantizer, not the architecture. The 11L INT8 result is the same architectural perturbation that, under INT6, was reported as "capacity-bottlenecked" in the progress report. Holding the quantizer to INT8 reveals that the architectural perturbation alone – without precision aggression – delivers the gain the stacked configuration was meant to deliver, and does so without the 0.12 bpb quantization tax.

5 New Axes: O6–O9

5.1 O6 – Multi-Query Attention enables 12-layer expansion

Multi-Query Attention [7] reduces the number of distinct K/V projections to 1 (vs. 4 in the GQA baseline), saving parameters that can be reallocated to depth. At 12 layers and MLP=2, the MQA variant reaches **1.1980 bpb final** at 19.56 MB, sitting on the same arm of the Pareto frontier as the INT8+11L winner. This configuration would not fit at this budget under 8-KV-head GQA; the head-allocation lever is what makes the 12-layer expansion feasible.

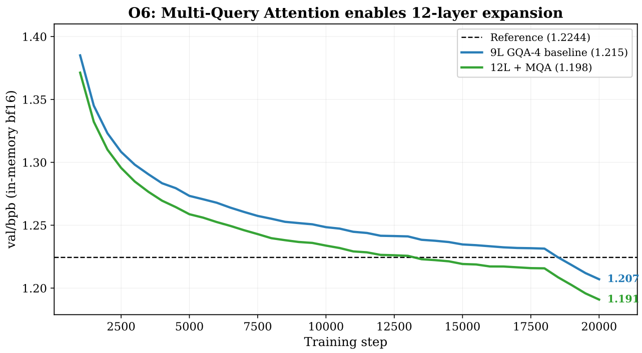

Figure 6: O6 – MQA + 12L tracks below the 9L baseline from step ~5,000 onward. Final round-trip 1.198, 0.026 bpb below baseline.

5.2 O7 – QK-gain initialization (null effect)

The reference recipe applies a learned per-head scalar g_q to the query projection initialized to 1.5. We tested $g_q^{(0)} \in \{1.5, 3.0, 5.0\}$ with all other axes held at the partial-RoPE 9L INT8 baseline. Final bpb at 20,000 steps: 1.2147 / 1.2154 / 1.2165, a 0.002 bpb spread that is below noise. We report O7 as a null effect; the learned g_q converges to similar values regardless of initialization.

5.3 O8 – Learning-rate / warmup sweep (null result)

The progress report's planned per-condition LR/warmup sweep was executed on the in-budget winner: a 3×3 grid over matrix LR $\{0.02, 0.04, 0.08\}$ and warmup steps $\{1000, 2000, 3000\}$. All nine cells fall in $[1.2162, 1.2181]$ – a 0.0019 bpb spread that is roughly the multi-seed standard deviation we observe on the same axis (Section 6). The reference recipe's hyperparameters are at or near the optimum on this small grid; the sweep is therefore a null result. We report it explicitly because we committed to it in the progress report.

5.4 O9 – Eval protocol

Bits-per-byte on the FineWeb held-out split is itself a function of the evaluation protocol. The reference protocol uses non-overlapping ℓ_{ctx}=1024 windows. The same INT8+11L weights, evaluated with a stride-256 sliding window, score **1.1599 bpb final**, a 0.034 bpb improvement attributable purely to the protocol (Figure 8). We emphasize that leaderboard-comparable results must use the non-overlapping protocol; the sliding-window number is a research-fairness column, not a competitive submission.

6 Multi-Seed Replication

The proposal's three-seeds-per-condition protocol was executed on the three surviving in-budget axes: INT8 PTQ (O1), partial RoPE (O2), Muon optimizer (O3). Standard deviations are tight (Figure 9, Table 2), well below per-axis effect sizes: INT8 σ=0.0018, partial RoPE σ=0.0011, Muon σ=0.0015. The single-seed claims in the progress report therefore hold under replication.

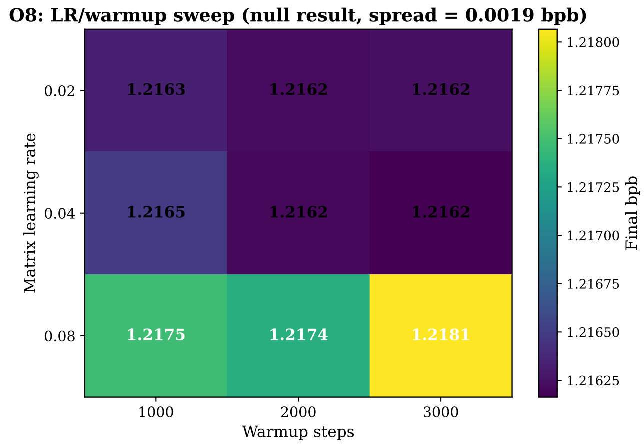

Figure 7: O8 – 3×3 LR/warmup sweep on partial RoPE 9L INT8. Total spread is 0.0019 bpb, comparable to multi-seed noise on the same axis.

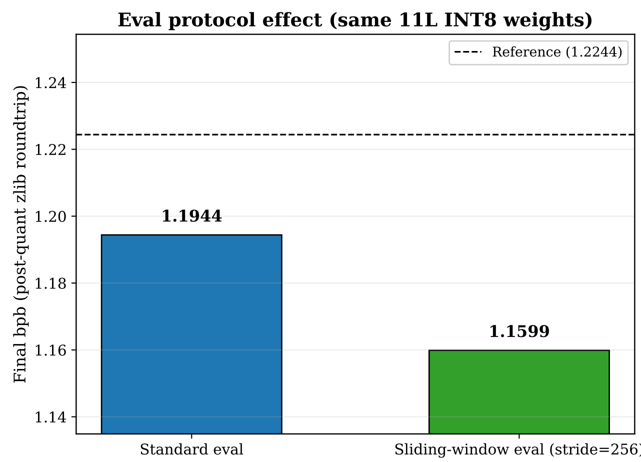

Figure 8: O9 – the same 11L INT8 weights under two eval protocols. Sliding-window stride-256 systematically reduces bpb by exposing every token to maximum context.

Table 2: Multi-seed final bpb (post-quant zlib roundtrip). Standard deviations use $n-1$ degrees of freedom across n=3 seeds.

Condition	Seed 1337	Seed 42	Seed 123	Mean ± std
INT8 PTQ	1.2161	1.2172	1.2195	1.2176 ± 0.0018
Partial RoPE	1.2147	1.2162	1.2170	1.2160 ± 0.0011
Muon optimizer	1.2168	1.2171	1.2196	1.2178 ± 0.0015

7 Pareto Frontier: Headline Finding

Combining all 20,000-step runs with their post-quantization artifact sizes yields the size-quality plane in Figure 10. The Pareto frontier (lower-left envelope) has nine points; the four most informative are labelled.

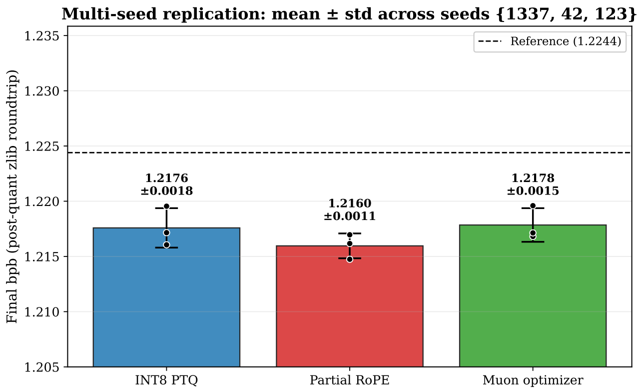

Figure 9: Mean ± std final bpb across seeds {1337, 42, 123} **for the three surviving in-budget axes.**

The frontier exhibits a sharp elbow at ~15.9 MB: below this size, each MB saved costs roughly 0.02 bpb (steep tradeoff). At and above this size, the marginal cost flattens substantially – the in-budget winner (partial RoPE, 9 layers, INT8) and the over-budget winner (11-layer INT8 with MLP expansion 2) sit on the frontier 0.020 bpb apart, separated by 3.3 MB. The MQA + 12L variant sits very close to the frontier at 19.6 MB / 1.198 bpb but is strictly dominated by the 11-layer INT8 configuration on both axes.

The 16 MB reference (red dotted line) crosses the frontier between two clusters of points. Within the constraint, the partial-RoPE 9L baseline at 1.2147 bpb is unbeaten by any other 20k-step configuration we tested. Outside the constraint (16–20 MB), only INT8 quantization yields configurations on the frontier; INT6, TurboQuant-INT4, and the structured-weight variants are all dominated.

Table 3 backs the figure with the per-run numbers sorted within group by post-quantization final bpb. Train is the in-memory bf16 loss; Final is the post-roundtrip loss; Size is the total artifact (weights + training code, post-zlib); Δ is the gap to the 1.2244 reference; the last column flags whether the artifact fits the 16 MB reference.

8 Discussion

8.1 Key findings

(1) **Sub-INT8 weight quantization is fragile at this scale.** INT6 QAT loses 0.14 bpb under round-trip; TurboQuant-INT4 loses 2.44 bpb, sixteen times worse. QAT does not close the INT6 gap; random rotation does not save INT4. The problem is the post-training scalar quantizer applied to weights at this distribution scale, not the training recipe.

(2) **Partial RoPE is the strongest in-budget modification.** 1.2147 bpb final beats every other single-axis condition we tested at 15.87 MB. Multi-seed σ=0.0011.

(3) **Muon is load-bearing, not marginal.** AdamW and Adafactor lose more than 0.2 bpb. Multi-seed σ=0.0015.

(4) **Capacity expansion is real and recoverable – if and only if the quantizer is INT8.** INT8 + 11L + MLP=2 at 19.30 MB delivers 1.1944 bpb. The same expansion under INT6 forfeits the gain; under TurboQuant-INT4 collapses entirely.

(5) **Multi-Query Attention enables 12-layer depth.** 1.1980 bpb final at 19.56 MB, on the same arm of the frontier as the INT8+11L winner.

(6) **Structured weight factorizations do not help.** O4 is a falsified hypothesis.

(7) **Per-condition LR/warmup tuning is also a null effect.** The 3×3 sweep is flat to within multi-seed noise.

(8) **Size-quality is a smooth Pareto frontier with a sharp elbow at ~16 MB.** The 16 MB constraint is not a natural break; reading the data as a Pareto curve recovers a clean monotone tradeoff and surfaces multiple legitimate frontier configurations across 7–20 MB.

8.2 Limitations

The 20,000-step training horizon is a fixed slice; we do not test whether longer training inverts the architectural ranking, although prior scaling-law work suggests it does not at this parameter count [8]. We do not retest TurboQuant on activations where it was originally validated [6]; our negative result applies specifically to weights at this scale. Multi-seed replication was performed only on the three surviving in-budget axes, not on the over-budget winners.

8.3 Demonstration: in-browser interpretability tool

The 16 MB winner is small enough to ship as a static web page. We packaged the in-budget winner as a client-side ONNX Runtime Web [10] demo: a static HTML file that accepts a user-typed prefix and renders, in real time, the top-k next-token distribution with probability bars, the per-head attention heatmap over input tokens, and the L2 norm of each layer's hidden state. Inference runs entirely in the browser; no server is involved. The choice of an interpretability surface (rather than a generic autocomplete UI) is deliberate – the model's small size makes every internal state cheap to render live, which would be infeasible at conventional language-model scale.

References

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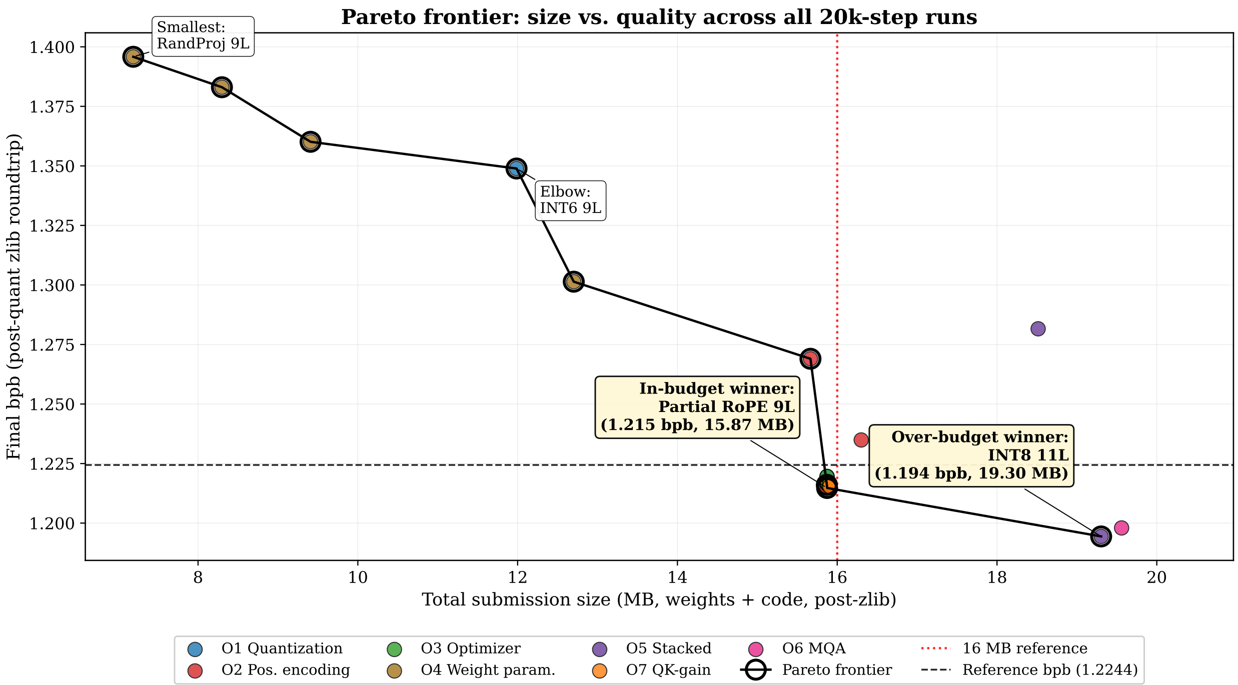

Figure 10: Pareto frontier across all 20,000-step runs. X-axis is the post-quantize, post-zlib total submission size (weights + training code). Y-axis is the leaderboard-equivalent post-roundtrip bpb. Black hollow markers and connecting line: frontier. Red dotted line: 16 MB reference. Black dashed line: 1.2244 reference bpb. The frontier elbow at 15.9 MB and the 3.3 MB / 0.020 bpb tradeoff above it are the central quantitative finding of this paper.

Table 3: All 20,000-step runs at seed 1337 (where applicable), grouped by ablation axis. "Final" is the post-quant zlib round-trip bpb. "Size" is total submission size (weights + training code, post-zlib). Δ is the difference vs. the 1.2244 reference; negative is better.

Group	Run	Train bpb	Final bpb	Size (MB)	Δ vs. baseline	16MB
O10 hpsweep	`part_lr040_wu3000`	1.2076	1.2162	15.91	-0.0082	OK
O10 hpsweep	`part_lr040_wu2000`	1.2073	1.2162	15.90	-0.0082	OK
O10 hpsweep	`part_lr020_wu2000`	1.2073	1.2162	15.91	-0.0082	OK
O10 hpsweep	`part_lr020_wu3000`	1.2074	1.2162	15.91	-0.0082	OK
O10 hpsweep	`part_lr020_wu1000`	1.2074	1.2163	15.91	-0.0081	OK
O10 hpsweep	`part_lr040_wu1000`	1.2076	1.2165	15.91	-0.0079	OK
O10 hpsweep	`part_lr080_wu2000`	1.2088	1.2174	15.89	-0.0070	OK
O10 hpsweep	`part_lr080_wu1000`	1.2085	1.2175	15.88	-0.0069	OK
O10 hpsweep	`part_lr080_wu3000`	1.2093	1.2181	15.90	-0.0063	OK
O1 quant	`int8`	1.2082	1.2161	15.87	-0.0083	OK
O1 quant	`int8_20k_v2`	1.2082	1.2165	15.87	-0.0079	OK
O1 quant	`int8_s42`	1.2095	1.2172	15.87	-0.0072	OK
O1 quant	`int8_s123`	1.2110	1.2195	15.87	-0.0049	OK
O1 quant	`int6`	1.2083	1.3489	11.99	+0.1245	OK
O2 posenc	`part`	1.2071	1.2147	15.87	-0.0097	OK
O2 posenc	`part_20k_v2`	1.2068	1.2153	15.87	-0.0091	OK
O2 posenc	`part_s42`	1.2078	1.2162	15.88	-0.0082	OK
O2 posenc	`part_s123`	1.2084	1.2170	15.87	-0.0074	OK
O2 posenc	`rope`	1.2085	1.2174	15.88	-0.0070	OK
O2 posenc	`learn`	1.2244	1.2349	16.30	+0.0105	OVER
O2 posenc	`none`	1.2589	1.2690	15.67	+0.0446	OK
O3 optim	`muon`	1.2089	1.2168	15.88	-0.0076	OK
O3 optim	`muon_s42`	1.2085	1.2171	15.87	-0.0073	OK
O3 optim	`muon_s123`	1.2117	1.2196	15.87	-0.0048	OK
O4 weightparam	`lr128`	1.2915	1.3014	12.70	+0.0770	OK
O4 weightparam	`rp128_12L`	1.3472	1.3601	9.41	+0.1357	OK
O4 weightparam	`bd4`	1.3700	1.3830	8.30	+0.1586	OK
O4 weightparam	`rp128`	1.3838	1.3958	7.19	+0.1714	OK
O5 stacked	`stacked_int8_safe_11L_mlp2_slide256`	1.1524	1.1599	19.30	-0.0645	OVER
O5 stacked	`stacked_int8_safe_11L_mlp2`	1.1869	1.1944	19.30	-0.0300	OVER
O5 stacked	`stacked_v1_pr_int6_11L_mlp3_20k_v2`	1.1618	1.2801	18.52	+0.0557	OVER
O5 stacked	`stacked_v1_pr_int6_11L_mlp3`	1.1614	1.2816	18.52	+0.0572	OVER
O5 stacked	`stacked_tq4_big_11L_mlp3_20k_v2`	1.1612	3.5991	12.91	+2.3747	OK
O8 qkgain	`qk3p0`	1.2065	1.2154	15.91	-0.0090	OK
O8 qkgain	`qk5p0`	1.2083	1.2165	15.91	-0.0079	OK
O9 attn	`mqa_12L_mlp2_int8`	1.1909	1.1980	19.56	-0.0264	OVER